

MH2500 Probability and Introduction to Statistics

Slides for Week 2 — 1

Topics: Counting Principles — Practice

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Exercise:

How many different 7-place licence plates are possible if the first 3 places are to be occupied by letters and the final 4 by numbers, and repetition among letters or numbers are prohibited?

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x	y	z	k	.	.	.

- x , 26 choices
- $y \neq x$, 25 choices
- $z \neq x, y$, 24 choices
- 10 choices for k
- ...

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(Answer: $26 \times 25 \times 24 \times 10 \times 9 \times 8 \times 7 = 78,624,000$)

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How many different letter arrangements can be formed from the letters *PEPPER*?

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We first note that there are $6!$ permutations of the letters $P_1 E_1 P_2 P_3 E_2 R$ when the $3P$'s and the $2E$'s are distinguished from each other.

Consider any one of these permutations — for instance,
 $P_1 P_2 E_1 P_3 E_2 R$.

If we now permute the P 's among themselves and the E 's among themselves, then the resultant arrangement would still be of the form $PPEPER$.

That is, all $3! \cdot 2!$ permutations

$P_1 P_2 E_1 P_3 E_2 R$ $P_1 P_2 E_2 P_3 E_1 R$ $P_1 P_3 E_1 P_2 E_2 R$ $P_1 P_3 E_2 P_2 E_1 R$
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are of the same form $PPEPER$.

Hence, there are $\frac{(6!)}{(3! \cdot 2!)}$ many possible letter arrangements of the letters $PEPPER$.

Exercise:

A chess tournament has 10 competitors, of which 4 are Russian, 3 are from the United States, 2 are from Great Britain, and 1 is from Brazil.

If the tournament result lists just the nationalities of the players in the order in which they placed, how many outcomes are possible?

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We first note that there are $10!$ permutations of 10 people with 10 nationalities $R_1, R_2, R_3, R_4, U_1, U_2, U_3, G_1, G_2$ and B .

- Each order is repeated by $4!3!2!1!$ times.

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Hence, there are $(10!)/(4! \cdot 3! \cdot 2!)$ many possible orders.

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Ten children are to be divided into an A team and a B team of 5 each. The A team will play in one league and the B team in another. How many different divisions are possible?

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(c) How many choices are there in which 2 of the 3 students are in the same class and the other student is in a different class?

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$$\binom{3n}{3} = \binom{3}{1} \binom{n}{3} + \binom{3}{1} \binom{n}{2} \binom{2}{1} \binom{n}{1} + \binom{n}{1} \binom{n}{1} \binom{n}{1}$$

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$$\binom{3n}{3} = 3 \binom{n}{3} + 3n^2(n-1) + n^3.$$

- Combinatorial Proof

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How many distinct nonnegative integer-valued vectors $(x_1, x_2, \dots, x_5)$ satisfying the equation

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Consider the following figure

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!!!!++!!!!++!

That is, $x_1 = 4, x_2 = 0, x_3 = 5, x_4 = 0$ and $x_5 = 1$ is one such solution.

Another way to think about it:

We have altogether $10 + 4$ positions, on a line. How many distinct ways of putting 4 many $+$ on these positions?

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It is the answer to [the question on last slide of week 1](#).

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which gives a solution $(1, 3, 2, 1, 3)$.

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which has 9 gaps between !s. We want to allocate 4 +s on these 9 gaps. For example, we can have the following allocation

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which gives a solution $(1, 3, 2, 1, 3)$.

In total, we have $\binom{10-1}{5-1} = \binom{9}{4}$ such allocations of 4 +s, and hence $\binom{9}{4}$ distinct positive solutions to the equation $x_1 + x_2 + \dots + x_5 = 10$.

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for which exactly k of the x_i are equal to 0.

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(Answer: $\binom{r}{k} \binom{n-1}{r-k-1}$)